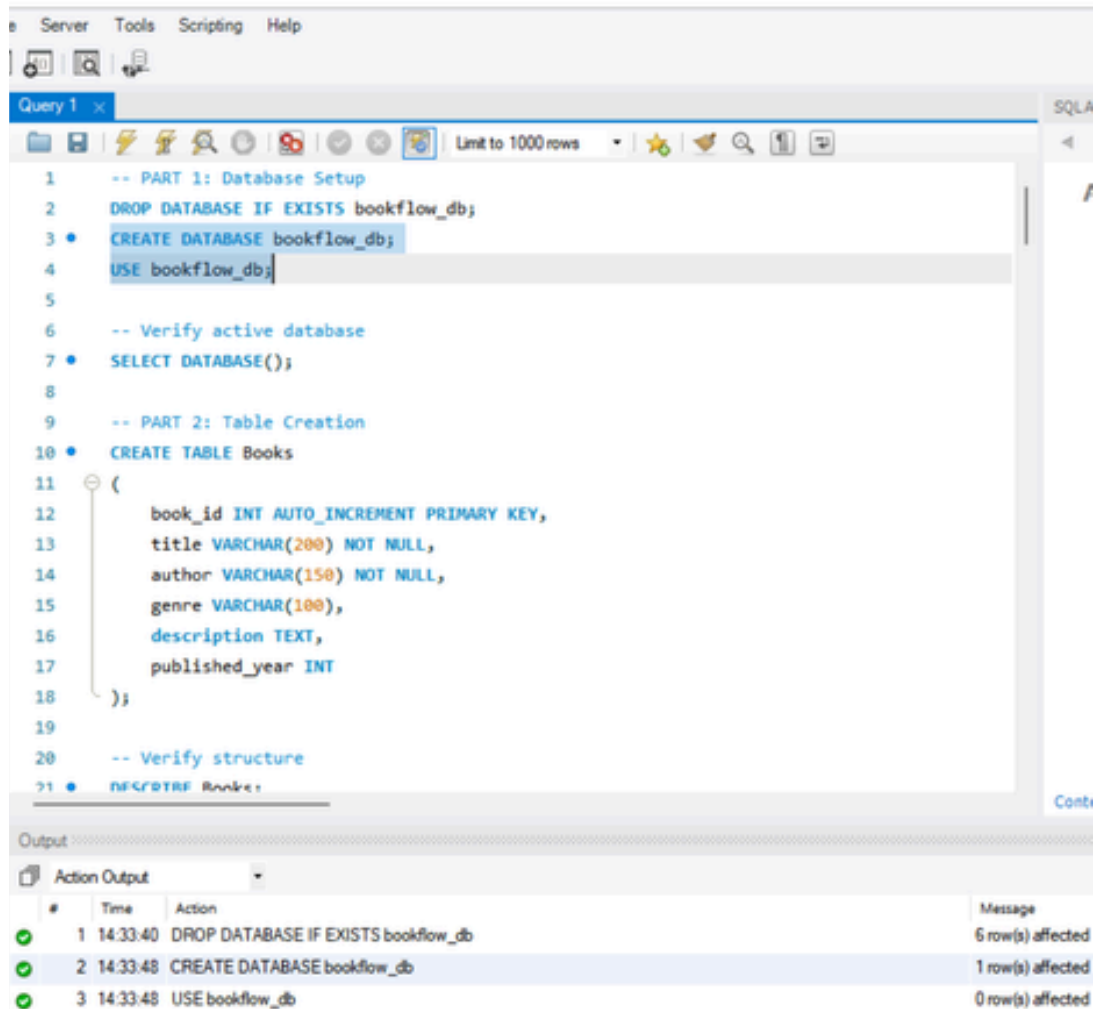


Practical Week 4

CREATE DATA BASE



```
1  -- PART 1: Database Setup
2  DROP DATABASE IF EXISTS bookflow_db;
3  CREATE DATABASE bookflow_db;
4  USE bookflow_db;
5
6  -- Verify active database
7  SELECT DATABASE();
8
9  -- PART 2: Table Creation
10 CREATE TABLE Books
11 (
12     book_id INT AUTO_INCREMENT PRIMARY KEY,
13     title VARCHAR(200) NOT NULL,
14     author VARCHAR(150) NOT NULL,
15     genre VARCHAR(100),
16     description TEXT,
17     published_year INT
18 );
19
20 -- Verify structure
21
```

Output

#	Time	Action	Message
✓ 1	14:33:40	DROP DATABASE IF EXISTS bookflow_db	6 row(s) affected
✓ 2	14:33:48	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:48	USE bookflow_db	0 row(s) affected

SQL Server Enterprise Edition

Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
1 -- PART 1: Database Setup
2 DROP DATABASE IF EXISTS bookflow_db;
3 CREATE DATABASE bookflow_db;
4 USE bookflow_db;
5
6 -- Verify active database
7 SELECT DATABASE();
8
9 -- PART 2: Table Creation
10 CREATE TABLE Books
11 (
12     book_id INT AUTO_INCREMENT PRIMARY KEY,
13     title VARCHAR(200) NOT NULL,
14     author VARCHAR(200) NOT NULL
15 );
```

Result Grid

Result Grid
DATABASE()
bookflow_db

Result 1 x

Read Only

Output

Action Output

#	Time	Action	Message
1	14:33:40	DROP DATABASE IF EXISTS bookflow_db	6 row(s) affected
2	14:33:48	CREATE DATABASE bookflow_db	1 row(s) affected
3	14:33:48	USE bookflow_db	0 row(s) affected
4	14:34:09	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned

TABLE CREATION

Server Tools Scripting Help

Query 1 x SQL

Limit to 1000 rows

```

1  -- PART 1: Database Setup
2  DROP DATABASE IF EXISTS bookflow_db;
3  • CREATE DATABASE bookflow_db;
4  USE bookflow_db;
5
6  -- Verify active database
7  • SELECT DATABASE();
8
9  -- PART 2: Table Creation
10 • CREATE TABLE Books
11 (
12     book_id INT AUTO_INCREMENT PRIMARY KEY,
13     title VARCHAR(200) NOT NULL,
14     author VARCHAR(150) NOT NULL,
15     genre VARCHAR(100),
16     description TEXT,
17     published_year INT
18 );
19
20 -- Verify structure
21 • DESCRIBE Books;

```

Output

Action Output

#	Time	Action	Message
✓ 1	14:33:40	DROP DATABASE IF EXISTS bookflow_db	6 row(s) affected
✓ 2	14:33:48	CREATE DATABASE bookflow_db	1 row(s) affected
✓ 3	14:33:48	USE bookflow_db	0 row(s) affected
✓ 4	14:34:09	SELECT DATABASE() LIMIT 0, 1000	1 row(s) returned
✗ 5	14:35:11	CREATE TABLE Books	Error Code: 1113.
✓ 6	14:35:22	CREATE TABLE Books (book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(200) NOT NU...	0 row(s) affected

STRUCTURE

Server Tools Scripting Help

Query 1

Limit to 1000 rows

```

19
20 -- Verify structure
21 • DESCRIBE Books;
22
23 -- PART 3: Data Insertion
24 • INSERT INTO Books
25 (
26     title,
27     author,
28     genre,
29     description,
30     published_year
31 )
32 VALUES
33 (
34     'The Martian',

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	NULL	auto_increment
	title	varchar(200)	NO		NULL	
	author	varchar(150)	NO		NULL	
	genre	varchar(100)	YES		NULL	
	description	text	YES		NULL	
	published_year	int	YES		NULL	

DATA INSERT

Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
23 -- PART 3: Data Insertion
24 INSERT INTO Books
25 (
26     title,
27     author,
28     genre,
29     description,
30     published_year
31 )
32 VALUES
33 (
34     'The Martian',
35     'Andy Weir',
36     'Science Fiction',
37     'An astronaut struggles to survive alone on Mars.',
38     2011
39 ),
40 (
41     'Dune',
42     'Frank Herbert',
43     'Science Fiction',
44     'A story involving planets, politics, survival and space.',
45     1965
46 ),
```

Output

Action Output

#	Time	Action
8	14:37:04	INSERT INTO Books (title, author, genre, description, published_year)
9	14:37:38	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian...

DISPLAY INSERT RECORDS

MySQL Workbench interface showing a query editor and a result grid.

Query Editor:

```
101      1870
102    );
103
104    -- Display inserted records
105    • SELECT
106        book_id,
107        title,
108        author,
109        genre,
110        published_year
111    FROM Books
112    ORDER BY book_id;
113
114    -- PART 4: Add Embedding Column (JSON format)
115    • ALTER TABLE Books
116        ADD COLUMN embedding JSON;
117
```

Result Grid:

	book_id	title	author	genre	published_year
▶	1	The Martian	Andy Weir	Science Fiction	2011
	2	Dune	Frank Herbert	Science Fiction	1965
	3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
	4	Pride and Prejudice	Jane Austen	Romance	1813
	5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
	6	Foundation	Isaac Asimov	Science Fiction	1951
	7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
	8	Enders Game	Orson Scott Card	Science Fiction	1985
	9	Harry Potter and the Sorcerers Stone	J.K. Rowling	Fantasy	1997
	10	Twenty Thousand Leagues Under the Seas	Jules Verne	Adventure	1870
•	NULL	NULL	NULL	NULL	NULL

ADDING COLUMN EMBEDDING

MySQL Workbench

Arjun x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
110     published_year
111 FROM Books
112 ORDER BY book_id;
113
114 -- PART 4: Add Embedding Column (JSON format)
115 • ALTER TABLE Books
116   ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
```

Output

Action Output

#	Time	Action
9	14:37:38	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian', 'Andy Weir', 'S
10	14:38:04	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1000
11	14:38:46	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1000
12	14:39:34	ALTER TABLE Books ADD COLUMN embedding JSON

MySQL Workbench interface showing a query editor and a result grid.

Query Editor:

```
110         published_year
111     FROM Books
112     ORDER BY book_id;
113
114     -- PART 4: Add Embedding Column (JSON format)
115 • ALTER TABLE Books
116     ADD COLUMN embedding JSON;
117
118     -- Verify column addition
119 • DESCRIBE Books;
120
121     -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.75, 0.05) WHERE book_id = 7;
```

Result Grid:

	Field	Type	Null	Key	Default	Extra
▶	book_id	int	NO	PRI	HULL	auto_increment
	title	varchar(200)	NO		HULL	
	author	varchar(150)	NO		HULL	
	genre	varchar(100)	YES		HULL	
	description	text	YES		HULL	
	published_year	int	YES		HULL	
	embedding	json	YES		HULL	

INSERT MOCK VECTOR DATA

MySQL Workbench

Arun x

File Edit View Query Database Server Tools Scripting Help

Query 1 x

Limit to 1000 rows

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output

Action Output

#	Time	Action
✓ 11	14:38:46	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1000
✓ 12	14:39:34	ALTER TABLE Books ADD COLUMN embedding JSON
✓ 13	14:39:59	DESCRIBE Books
✓ 14	14:40:28	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1

MySQL Workbench interface showing a SQL query in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 12	14:39:34	ALTER TABLE Books ADD COLUMN embedding JSON
✓ 13	14:39:59	DESCRIBE Books
✓ 14	14:40:28	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1
✓ 15	14:40:50	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2

MySQL Workbench interface showing a SQL query in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 13	14:39:59	DESCRIBE Books
✓ 14	14:40:28	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1
✓ 15	14:40:50	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2
✓ 16	14:41:05	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3

MySQL Workbench interface showing a query editor and output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output

Action Output

#	Time	Action
✓ 14	14:40:28	UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1
✓ 15	14:40:50	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2
✓ 16	14:41:05	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3
✓ 17	14:41:23	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4

MySQL Workbench interface showing a SQL query script in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Action Output

#	Time	Action
✓ 15	14:40:50	UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2
✓ 16	14:41:05	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3
✓ 17	14:41:23	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4
✓ 18	14:41:36	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5

MySQL Workbench interface showing a SQL query in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 16	14:41:05	UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3
✓ 17	14:41:23	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4
✓ 18	14:41:36	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5
✓ 19	14:41:51	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6

MySQL Workbench interface showing a SQL query in the Query Editor and its execution output in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 17	14:41:23	UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4
✓ 18	14:41:36	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5
✓ 19	14:41:51	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 20	14:42:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7

MySQL Workbench interface showing a SQL query in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 18	14:41:36	UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5
✓ 19	14:41:51	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 20	14:42:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7
✓ 21	14:42:24	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8

MySQL Workbench interface showing a query editor and output window.

Query 1

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output

Action Output

#	Time	Action
✓ 19	14:41:51	UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6
✓ 20	14:42:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7
✓ 21	14:42:24	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8
✓ 22	14:42:42	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9

MySQL Workbench interface showing a SQL query in the Query Editor and its execution results in the Output panel.

Query Editor:

```
116 ADD COLUMN embedding JSON;
117
118 -- Verify column addition
119 • DESCRIBE Books;
120
121 -- PART 5: Insert Mock Vector Data
122 • UPDATE Books SET embedding = JSON_ARRAY(0.90, 0.80, 0.20) WHERE book_id = 1;
123 • UPDATE Books SET embedding = JSON_ARRAY(0.85, 0.75, 0.25) WHERE book_id = 2;
124 • UPDATE Books SET embedding = JSON_ARRAY(0.80, 0.70, 0.30) WHERE book_id = 3;
125 • UPDATE Books SET embedding = JSON_ARRAY(0.10, 0.20, 0.90) WHERE book_id = 4;
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
```

Output Panel:

Output: Action Output

#	Time	Action
✓ 20	14:42:06	UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7
✓ 21	14:42:24	UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8
✓ 22	14:42:42	UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9
✓ 23	14:42:57	UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10

INSERTED ALL VECTORS

MySQL Workbench interface showing a query editor and a result grid.

Query Editor:

```
126 • UPDATE Books SET embedding = JSON_ARRAY(0.15, 0.30, 0.80) WHERE book_id = 5;
127 • UPDATE Books SET embedding = JSON_ARRAY(0.95, 0.65, 0.20) WHERE book_id = 6;
128 • UPDATE Books SET embedding = JSON_ARRAY(0.30, 0.75, 0.90) WHERE book_id = 7;
129 • UPDATE Books SET embedding = JSON_ARRAY(0.88, 0.85, 0.15) WHERE book_id = 8;
130 • UPDATE Books SET embedding = JSON_ARRAY(0.20, 0.55, 0.95) WHERE book_id = 9;
131 • UPDATE Books SET embedding = JSON_ARRAY(0.70, 0.90, 0.40) WHERE book_id = 10;
132
133 -- Verify all inserted vectors
134 • SELECT
135     book_id,
136     title,
137     embedding
138 FROM Books
139 ORDER BY book_id;
140
141 -- PART 6: L2 Distance Search Query
142 -- Search Vector: [0.90, 0.80, 0.20] (Stories about space)
143 • SELECT
```

Result Grid:

	book_id	title	embedding
▶	1	The Martian	[0.90, 0.80, 0.20]
	2	Dune	[0.85, 0.75, 0.25]
	3	The Hitchhikers Guide to the Galaxy	[0.80, 0.70, 0.30]
	4	Pride and Prejudice	[0.10, 0.20, 0.90]
	5	The Great Gatsby	[0.15, 0.30, 0.80]
	6	Foundation	[0.95, 0.65, 0.20]
	7	The Hobbit	[0.30, 0.75, 0.90]
	8	Enders Game	[0.88, 0.85, 0.15]
	9	Harry Potter and the Sorcerers Stone	[0.20, 0.55, 0.95]
	10	Twenty Thousand Leagues Under the Seas	[0.70, 0.90, 0.40]
•	NULL	NULL	NULL

DISTANCE SEARCH QUERY

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Query 1

Limit to 1000 rows

```
137     embedding
138 FROM Books
139 ORDER BY book_id;
140
141 -- PART 6: L2 Distance Search Query
142 -- Search Vector: [0.90, 0.80, 0.20] (Stories about space)
143 • SELECT
144     book_id,
145     title,
146     author,
147     genre,
148     ROUND(
149         Sqrt(
150             POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90, 2)
151             +
152             POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80, 2)
153             +
154             POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20, 2)
```

Result Grid

	book_id	title	author	genre	l2_distance
▶	1	The Martian	Andy Weir	Science Fiction	0
	8	Enders Game	Orson Scott Card	Science Fiction	0.0735
	2	Dune	Frank Herbert	Science Fiction	0.0866

COSINE SIMILARITY SEARCH QUERY

MySQL Workbench

File Edit View Query Database Server Tools Scripting Help

Query 1

Limit to 1000 rows

```
163 -- PART 7: Cosine Similarity Search Query
164 -- Search Vector: [0.90, 0.80, 0.20] (Stories about space)
165 • SELECT
166     book_id,
167     title,
168     author,
169     genre,
170     ROUND(
171         (
172             CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) * 0.90
173         +
174             CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) * 0.80
175         +
176             CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) * 0.20
177         )
178         /
179         (
180             Sqrt(
```

Result Grid

	book_id	title	author	genre	cosine_similarity
▶	1	The Martian	Andy Weir	Science Fiction	1
	2	Dune	Frank Herbert	Science Fiction	0.9986
	8	Enders Game	Orson Scott Card	Science Fiction	0.9983